

HOMEWORK

Your Bike, Part 3

Four questions about architecture, asked of the same bike

Name: _____

Date: _____

Reading: Chapter 6 of E. Crawley, B. Cameron, and D. Selva (2015). *System Architecture: Strategy and Product Development for Complex Systems*. Upper Saddle River, NJ: Prentice Hall.

The Bike

This is the same bike as in the in-class activity on form and the homework on function. It has a steel frame, one gear, brakes that squeeze the wheel rims, and a rack over the back wheel. It also has a bell and reflectors. The rear axle is clamped into two slots at the back of the frame. You ride it to campus every day.

What You Are Doing

The in-class activity described what the bike **is**, and the homework on function described what it **does**. Chapter 6 puts the two together: it asks **which part of the bike does what**, and how the parts have to be arranged for the bike to work. The chapter organizes this into seven questions, 6a through 6g, collected in its Table 6.1. The four parts below cover the first four, 6a through 6d. Every part opens with a worked example from the text. Copy its pattern. Short answers are fine, and a sketch is often better than a paragraph.

Reminder: the vocabulary of Chapter 6

System architecture is “the embodiment of concept, the allocation of physical/informational function to the elements of form, and the definition of relationships among the elements and with the surrounding context” (Box 6.1).

Architecture is not a third thing on top of form and function. It is the **mapping** between them (Figure 6.1). In OPM, that mapping is the line ending in a small circle that joins each **instrument** (an element of form) to the **process** it makes possible.

Given: the pedaling pathway from Part 2

leg force → **Cranking** → crank torque → **Pulling** → chain tension → **Turning** → wheel torque → **Pushing** → forward force at the road → **Moving** → rider (location)

Each process consumes the operand before it and creates the one after it. **Moving** is the exception: it affects the rider by changing the rider's location.

1 Which part does each process?**Worked example (Figure 6.8 and Table 6.2 of the text)**

For the pump, the text goes process by process and asks what element of form makes each one possible. The **impeller** is the instrument of accelerating. The **housing** is the instrument of both diffusing and outflowing, so the text warns that “the design of this part must skillfully blend these two functions.”

Table 6.2 records the same thing as a process-form (PF) array. Each row is a process and each column an element of form. An **I** in a cell means that part is an instrument of that process.

Your turn.

1a. Fill in the PF array for the pedaling pathway. A part can be the instrument of more than one process, and a process can need more than one instrument.

	Pedals and cranks	Chainring	Chain	Rear sprocket	Rear wheel	Frame	Saddle
Cranking							
Pulling							
Turning							
Pushing							
Moving							

1b. Find a part in your array that is the instrument of more than one process. Which part of the pump plays the same role? Then find a process that needs more than one instrument.

1c. In Figure 6.5b, the walker is both the operand and the instrument of walking: people “walk themselves.” One object in the pedaling pathway is an operand of the bike. Which one is it, and which processes is that same object an instrument of?

2 How does the structure carry the interactions?

Worked example (Section 6.2 of the text)

In the pump, the inflowing process (instrument: the cover) passes the internal low-pressure flow to the accelerating process (instrument: the impeller). So something in the form has to let the flow get from the cover to the impeller. The text finds that the two are not connected at all. The only relationship between them is spatial: the cover is “close to” the impeller. That is enough, the way a garden hose pointed at a bucket fills the bucket without touching it. And the size of the gap *informs performance*: a smaller gap gives a higher rise in pressure.

The general rule: connections usually **enable** a functional interaction, while location usually **informs** how well it performs.

Your turn.

2a. Each functional interaction from Part 2 has to get from the instrument of one process to the instrument of the next. For each one, name the formal relationship that lets it pass, and mark it **C** for a connection or **S** for a spatial relationship, as in the in-class activity.

Interaction	Passed from	Passed to	Formal relationship	C or S
Crank torque	Cranks	Chainring		
Chain tension	Chain	Rear sprocket		
Wheel torque	Rear sprocket	Rear rim		
Forward force	Rear wheel	Frame		

2b. The forward force is created where the tire meets the road. Nothing bolts, meshes, or clamps the tire to the road. What formal relationship lets the force pass there, and which relationship in the pump does it resemble?

2c. Two spatial relationships matter a great deal on a one-gear bike. The chainring and the rear sprocket have to lie in the same plane. And the position of the rear axle in its slots sets how tight the chain is. For each one, does it *enable* the interaction or *inform* its performance? What happens to the ride as it drifts out of adjustment?

3 What else is on the bike?

Worked example (Section 6.3 and Figure 6.10 of the text)

After the pump's value pathway is mapped, the seal, O-ring, motor, motor shaft, screws, locking nut, and mounting are still unassigned. The text sorts them into two kinds.

Non-idealities (Question 6b) handle the messy parts of the value pathway. Some manage the operand, by moving, containing, or storing it: the O-ring and the seal keep the water in. Others add performance or robustness.

Supporting functions (Question 6c) hold up, power, and control the instruments of the value pathway. The motor drives and supports the shaft, the shaft drives and supports the impeller, and the mounting supports the motor. Figure 6.10 draws these as extra columns of processes and instruments to the right of the value instruments.

Your turn.

3a. None of these four parts appear in your array from Part 1. Say whether each one is a non-ideality or a supporting function, and write its function as a process plus an operand.

Part	Non-ideality or supporting?	Process + operand
Chain oil		
Fenders		
Wheel bearings		
Kickstand		

3b. Draw two supporting layers to the right of the value pathway, as Figure 6.10 does. The first two columns of each row are given. In each pair of blanks, write a supporting process and the instrument that performs it, so that each instrument supports the one to its left.

Value process	Value instrument	Supporting process	Supporting instrument	Supporting process	Supporting instrument
Turning	Rear sprocket				
Cranking	Pedals and cranks				

Which part ends up in your last column, and where else in this homework have you already used it? The text has a remark about this for the pump's housing, at the end of the discussion of Figure 6.10. What does it say to do?

4 What happens at the boundary?

Worked example (Figures 6.11 and 6.12 of the text)

Every interface has form and function. The text says an interface definition must name:

- the **operand** that passes through the boundary, the same on both sides;
- the **process** of passing it, shared by both sides;
- the two **interface instruments**, one on each side, and how they are related. They are *androgynous* if they are identical, and *compatible* if they are different but fit together.

For the pump's inlet, the operand is water, the process is flowing, and the instruments are the inflow hose and the fitting on the pump's cover. For its power, the operand is electrical current, the process is supplying, and the instruments are a plug and a socket.

Your turn.

4a. Fill in the same four entries for three interfaces of your bike.

Interface	Operand	Process	The two instruments	How they are related
Foot on pedal				
Tire on road				
Pump on tire valve				

4b. Your tires have *Presta* valves, the thin kind with a small nut on the tip. Your roommate's pump has a head for *Schrader* valves, the kind on car tires. Both are standard, nothing is broken, and the pump will not go on. Which of the entries in your 4a row is the same on both sides, and which one failed? Are the valve and pump head androgynous or compatible? A two-dollar adapter fixes the problem. Where does the adapter sit in Figure 6.11?